Introduction to Gaussian Mixture Models and EM Algorithm

Motivation

- Real-world data often comes from multiple subpopulations.
- A single Gaussian distribution may not model complex data well.
- Gaussian Mixture Models (GMMs) represent data as a mixture of several Gaussian distributions.
- Widely used in:
    - Clustering
    - Density estimation
    - Speech recognition
    - Image segmentation

Gaussian Distribution

A 1D Gaussian distribution is defined as:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where:

- μ = mean
- σ^2 = variance

Multivariate Gaussian:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mu)^T\Sigma^{-1}(\mathbf{x}-\mu)\right)$$

What is a Gaussian Mixture Model?

A Gaussian Mixture Model assumes data is generated from a mixture of K Gaussian components.

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

where:

- π_k = mixing coefficient
- $\sum_{k=1}^{K} \pi_k = 1$
- $\mathcal{N}(x \mid \mu_k, \Sigma_k)$ = Gaussian component

Interpretation:

- Each Gaussian represents one cluster.
- The model gives soft cluster assignments.

Latent Variables

Introduce hidden variable:

$$z \in \{1, 2, \ldots, K\}$$

where:

- $z = k$ indicates sample belongs to component k

Joint probability:

$$p(x, z = k) = \pi_k \mathcal{N}(x|\mu_k, \Sigma_k)$$

Marginal probability:

$$p(x) = \sum_{k=1}^{K} p(x, z = k)$$

Why Maximum Likelihood is Difficult

Dataset: $X = \{x_1, x_2, \ldots, x_N\}$
Likelihood: $L(\theta) = \prod_{n=1}^{N} \sum_{k=1}^{K} \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k)$
Log-likelihood: $\log L(\theta) = \sum_{n=1}^{N} \log \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k) \right)$
Problem:

- Log of sum makes direct optimization difficult.

Solution:

- Expectation Maximization (EM) Algorithm

Expectation Maximization (EM)

EM is an iterative algorithm for maximum likelihood estimation with hidden variables.

Two steps repeated until convergence:

1. **E-step (Expectation)**
2. **M-step (Maximization)**

Goal:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta, \theta^{(t)})$$

where:

$$Q(\theta, \theta^{(t)}) = \mathbb{E}_{Z|X,\theta^{(t)}}[\log p(X, Z|\theta)]$$

E-Step: Compute Responsibilities

Compute posterior probability that component k generated sample x_n.

$$\gamma(z_{nk}) = p(z_k = 1 | x_n)$$

Using Bayes rule:

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_n | \mu_j, \Sigma_j)}$$

Interpretation:

- Soft assignment of points to clusters.

M-Step: Update Parameters

Using responsibilities, update parameters.
Effective number of points:

$$N_k = \sum_{n=1}^{N} \gamma(z_{nk})$$

Update mean:

$$\mu_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) x_n$$

Update covariance:

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk})(x_n - \mu_k)(x_n - \mu_k)^T$$

Update mixing coefficient:

EM Algorithm Summary

1. Initialize parameters:

$$\mu_k, \Sigma_k, \pi_k$$

2. Repeat until convergence:
    1. E-step: compute responsibilities
    2. M-step: update parameters
3. Stop when log-likelihood converges

Properties:

- Likelihood never decreases
- May converge to local optimum

GMM vs K-Means

Feature	K-Means	GMM
Cluster Shape	Spherical	Elliptical
Assignments	Hard	Soft
Covariance	Same	Different
Probability Model	No	Yes
Optimization	Distance	Likelihood

Key Insight:

- K-Means is a special case of GMM with:
  - Equal covariance matrices
  - Hard assignments

Applications of GMMs

- Speaker recognition
- Image segmentation
- Anomaly detection
- Recommendation systems
- Financial modeling
- Density estimation

Advantages:

- Flexible clustering
- Probabilistic interpretation

Limitations:

- Sensitive to initialization
- Computationally expensive

Conclusion

- GMM models data using multiple Gaussian distributions.
- EM algorithm estimates parameters efficiently.
- EM alternates between:
  - Estimating hidden variables
  - Updating parameters
- GMM provides probabilistic clustering unlike K-Means.

$$\boxed{\text{GMM} + \text{EM} = \text{Powerful Probabilistic Clustering}}$$

Step-by-Step EM Algorithm Example for 1D Gaussian Mixture Models

Problem Setup

We are given 1D data:

$$X = \{1,\ 2,\ 3,\ 10,\ 11,\ 12\}$$

Goal:

- Fit a Gaussian Mixture Model (GMM)
- Use:
  - $K = 2$ Gaussian components
  - EM algorithm

Assume:

$$\sigma_1^2 = \sigma_2^2 = 1$$

Initial parameters:

$$\mu_1^{(0)} = 2 \qquad \mu_2^{(0)} = 11$$

$$\pi_1^{(0)} = 0.5 \qquad \pi_2^{(0)} = 0.5$$

Gaussian Density Formula

For 1D Gaussian:

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Since:

$$\sigma^2 = 1$$

we get:

$$\mathcal{N}(x|\mu,1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2}\right)$$

Step 1: E-Step

Compute responsibilities:

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(x_n|\mu_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_n|\mu_j)}$$

Interpretation:

- Probability that point x_n belongs to cluster k

We compute responsibilities for each data point.

E-Step Example for $x = 1$

Compute Gaussian values.
For Cluster 1:

$$\mathcal{N}(1|2,1) = \frac{1}{\sqrt{2\pi}} e^{-(1-2)^2/2}$$

$$= 0.3989 \times e^{-0.5} \approx 0.2420$$

For Cluster 2:

$$\mathcal{N}(1|11,1) = 0.3989 \times e^{-50} \approx 0$$

Now compute responsibility:

$$\gamma(z_{1,1}) = \frac{0.5(0.2420)}{0.5(0.2420) + 0} \approx 1$$

$$\gamma(z_{1,2}) \approx 0$$

Thus:

Responsibilities for All Points

Data Point	$\gamma(z_{n1})$	$\gamma(z_{n2})$
1	1.000	0.000
2	1.000	0.000
3	1.000	0.000
10	0.000	1.000
11	0.000	1.000
12	0.000	1.000

Observation:

- Left points belong to Cluster 1
- Right points belong to Cluster 2

Step 2: M-Step

Update means:

$$\mu_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk}) x_n}{\sum_{n=1}^{N} \gamma(z_{nk})}$$

Update mixing coefficients:

$$\pi_k = \frac{N_k}{N}$$

where:

$$N_k = \sum_{n=1}^{N} \gamma(z_{nk})$$

Update Mean of Cluster 1

Responsibilities:

$$\gamma(z_{n1}) = [1,\ 1,\ 1,\ 0,\ 0,\ 0]$$

Compute:

$$N_1 = 1 + 1 + 1 = 3$$

Now update mean:

$$\mu_1 = \frac{1(1) + 1(2) + 1(3)}{3}$$

$$= \frac{6}{3} = 2$$

Thus:

$$\boxed{\mu_1^{new} = 2}$$

Update Mean of Cluster 2

Responsibilities:

$$\gamma(z_{n2}) = [0,\ 0,\ 0,\ 1,\ 1,\ 1]$$

Compute:

$$N_2 = 3$$

Update mean:

$$\mu_2 = \frac{10 + 11 + 12}{3}$$

$$= 11$$

Thus:

$$\boxed{\mu_2^{new} = 11}$$

Update Mixing Coefficients

Total number of samples:

$$N = 6$$

Update mixing coefficients:

$$\pi_1 = \frac{3}{6} = 0.5$$

$$\pi_2 = \frac{3}{6} = 0.5$$

Updated parameters:

$$\mu_1 = 2 \qquad \mu_2 = 11$$

$$\pi_1 = 0.5 \qquad \pi_2 = 0.5$$

What Happened?

EM algorithm performed:

1. E-Step:
    - Computed probabilities of cluster membership
2. M-Step:
    - Updated cluster parameters

In this example:

- Clusters are well separated
- Responsibilities became nearly binary

In real datasets:

- Responsibilities are soft probabilities
- Multiple EM iterations are needed

EM Algorithm Summary

Repeat until convergence:

1. Initialize parameters

$$\mu_k, \sigma_k^2, \pi_k$$

2. E-Step:

$$\gamma(z_{nk}) = p(z_k|x_n)$$

3. M-Step:
    - Update means
    - Update variances
    - Update mixing coefficients
4. Stop when likelihood converges

Key Insights

- GMM performs probabilistic clustering
- EM alternates between:
  - Estimating hidden variables
  - Updating parameters
- Unlike K-Means:
  - Assignments are soft
  - Clusters can have different variances

EM maximizes likelihood iteratively

References

- Christopher Bishop, *Pattern Recognition and Machine Learning*
- Kevin Murphy, *Machine Learning: A Probabilistic Perspective*
- Hastie, Tibshirani, Friedman, *Elements of Statistical Learning*